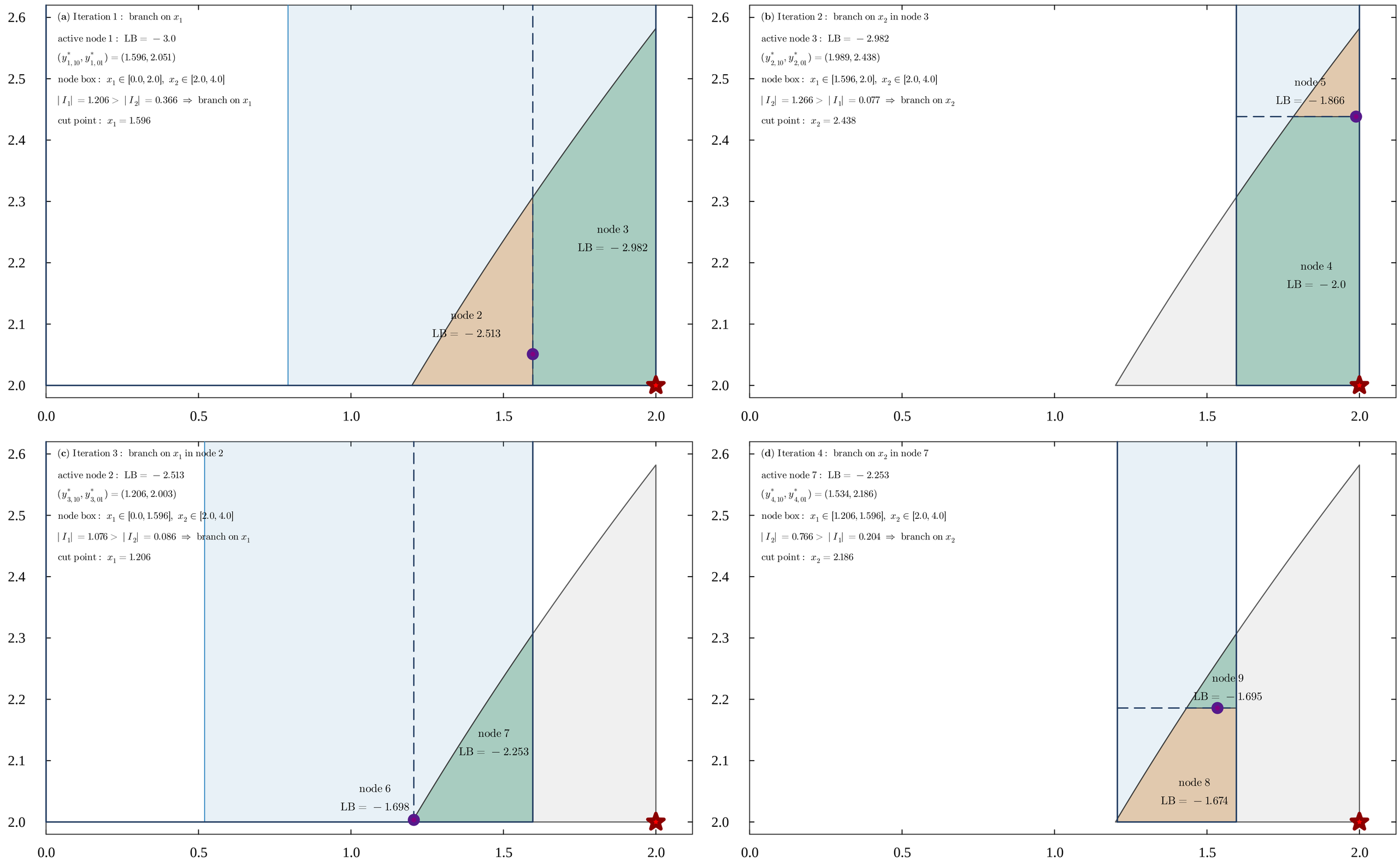
(a) Iteration 1 : branch on x_1
active node 1 : LB = -3.0
$(y^*_{1,10}, y^*_{1,01}) = (1.596, 2.051)$
node box : $x_1 \in [0.0, 2.0]$, $x_2 \in [2.0, 4.0]$
$|I_1| = 1.206 > |I_2| = 0.366 \Rightarrow$ branch on x_1
cut point : $x_1 = 1.596$
node 2
LB = -2.513
node 3
LB = -2.982
(b) Iteration 2 : branch on x_2 in node 3
active node 3 : LB = -2.982
$(y^*_{2,10}, y^*_{2,01}) = (1.989, 2.438)$
node box : $x_1 \in [1.596, 2.0]$, $x_2 \in [2.0, 4.0]$
$|I_2| = 1.266 > |I_1| = 0.077 \Rightarrow$ branch on x_2
cut point : $x_2 = 2.438$
node 5
LB = -1.866
node 4
LB = -2.0
(c) Iteration 3 : branch on x_1 in node 2
active node 2 : LB = -2.513
$(y^*_{3,10}, y^*_{3,01}) = (1.206, 2.003)$
node box : $x_1 \in [0.0, 1.596]$, $x_2 \in [2.0, 4.0]$
$|I_1| = 1.076 > |I_2| = 0.086 \Rightarrow$ branch on x_1
cut point : $x_1 = 1.206$
node 6
LB = -1.698
node 7
LB = -2.253
(d) Iteration 4 : branch on x_2 in node 7
active node 7 : LB = -2.253
$(y^*_{4,10}, y^*_{4,01}) = (1.534, 2.186)$
node box : $x_1 \in [1.206, 1.596]$, $x_2 \in [2.0, 4.0]$
$|I_2| = 0.766 > |I_1| = 0.204 \Rightarrow$ branch on x_2
cut point : $x_2 = 2.186$
node 9
LB = -1.695
node 8
LB = -1.674
2.6
2.5
2.4
2.3
2.2
2.1
2.0
0.0
0.5
1.0
1.5
2.0